

V3 moneyness performance

Same 6 models · 3 companies · 4 windows · 1.5-year monthly study · classified by call moneyness $m = S/K$

What this report is

- Companion to V3_1p5y_monthly_modified_gbm_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets ($n = 0$) show em dashes and have no BEST.

Unchanged experimental design

- Same one model, four companies (SPY, AAPL, MSFT, AMZN), 4 windows, 1.5-year lookback, monthly recalibration, V3 filters, Euler-corrected Heston step, LSM $n_{\text{paths}} = 10000$ / seed 42.
- Same option contracts as the main study (nearest-ATM listed calls, DTE 7-60, $|S/K - 1| \leq 10\%$). No model-specific re-sampling.
- Pooled $n = 802$ contracts per model (identical keys across models).

Moneyness rule (calls, $m = S / K$)

- DOTM: $m < 0.98$. OTM: $0.98 \leq m < 0.995$. ATM: $0.995 \leq m \leq 1.005$. ITM: $1.005 < m \leq 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \leq 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008-2009.

Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})^2)}$. Ranking uses this only.
- MAE and Bias \$ in option-premium dollars. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model $\times$ Moneyness RMSE% heatmap.
- Analysis 2 — Window $\times$ moneyness: keep the 4 windows separate, pool the four companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each window.

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2008-2009 Crisis	18	43	80	43	16	200
2013-2014 Normal	0	5	192	3	0	200
2018-2019 Late-cycle	0	4	191	7	0	202
2019-2020 COVID	0	6	189	5	0	200
All regimes	18	58	652	58	16	802

Moneyness rule · American calls, $m = S/K$

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	18	2.2%
OTM	$0.98 \leq m < 0.995$	58	7.2%
ATM	$0.995 \leq m \leq 1.005$	652	81.3%
ITM	$1.005 < m \leq 1.02$	58	7.2%
DITM	$m > 1.02$	16	2.0%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1-M3

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

Table M1 · DOTM · $m < 0.98$ · $n = 18$ · BEST = Modified GBM

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
Modified GBM	18	43.64	1.0624	+0.2588	+18.53	21.5%	1

Table M2 · OTM · $0.98 \leq m < 0.995$ · $n = 58$ · BEST = Modified GBM

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
Modified GBM	58	58.42	1.5461	+0.6604	+29.31	22.6%	1

Table M3 · ATM · $0.995 \leq m \leq 1.005$ · $n = 652$ · BEST = Modified GBM

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
Modified GBM	652	46.30	5.4180	+2.9257	+27.92	24.6%	1

Bias \$ = mean(model – market). Bias% = $100 \times \text{mean}((\text{model} - \text{market})/\text{market})$.

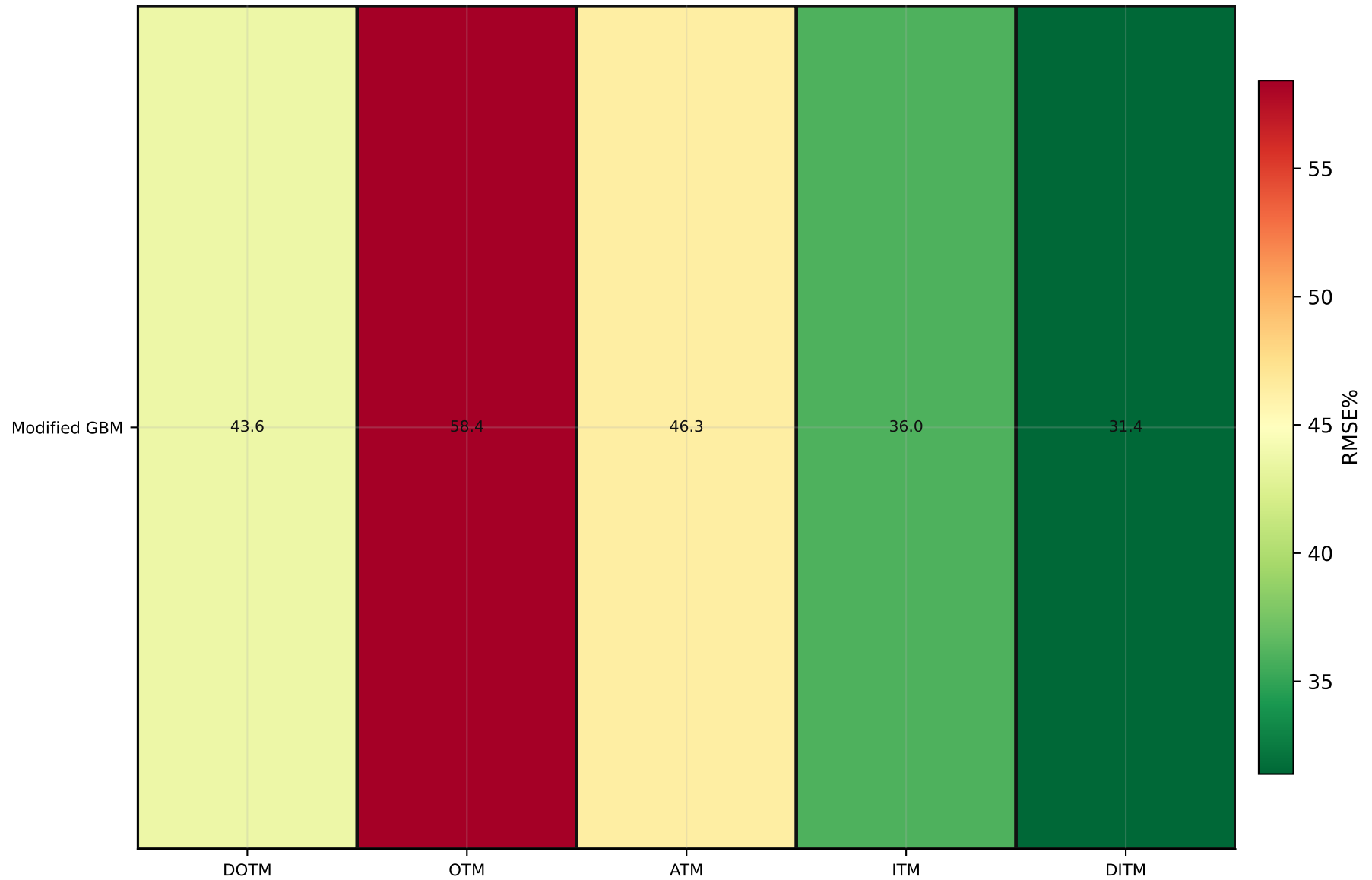
Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

Table M4 · ITM · $1.005 < m \leq 1.02$ · $n = 58$ · BEST = Modified GBM

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
Modified GBM	58	36.00	1.3075	+0.2659	+10.23	27.4%	1

Table M5 · DITM · $m > 1.02$ · $n = 16$ · BEST = Modified GBM

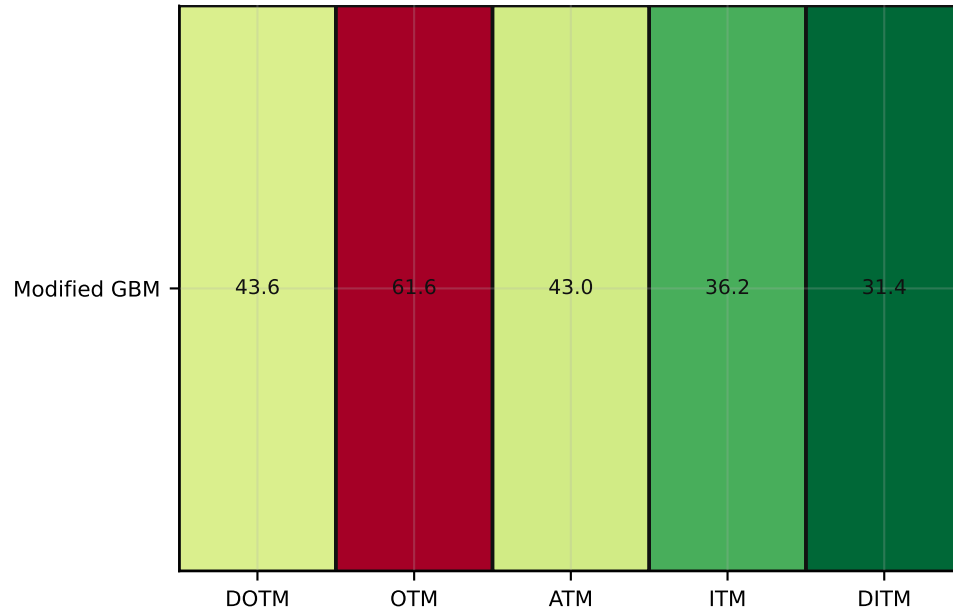
Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
Modified GBM	16	31.39	1.1988	+0.1199	+10.29	28.3%	1



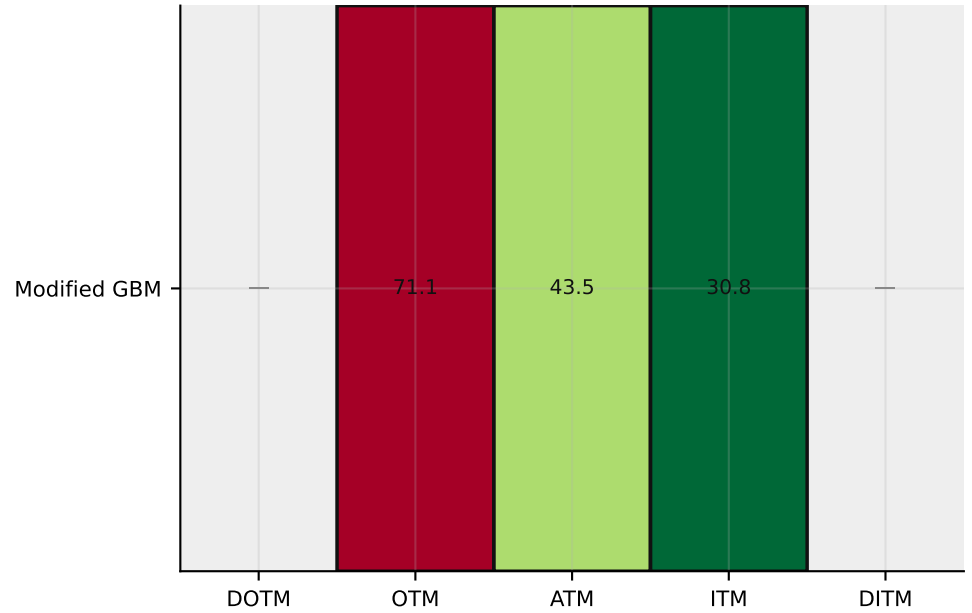
Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

Figure M2 · RMSE% by model × moneyiness · each window (companies pooled)

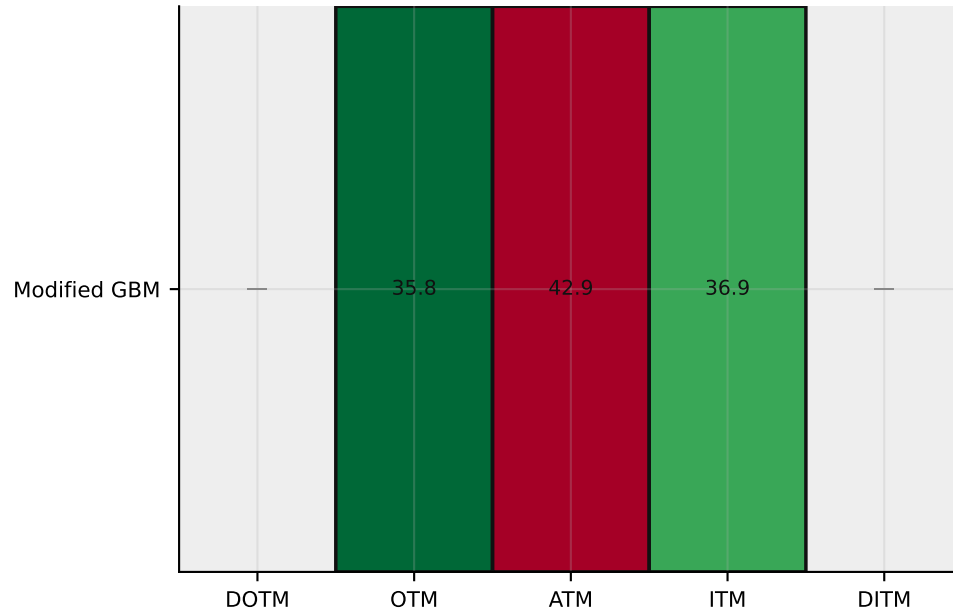
2008-2009 · Crisis



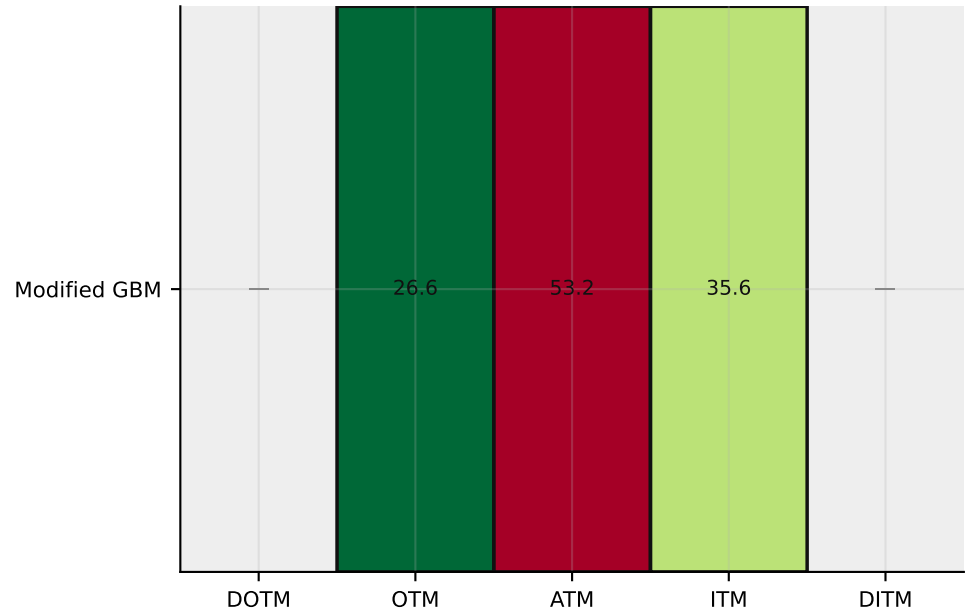
2013-2014 · Normal



2018-2019 · Late-cycle



2019-2020 · COVID



Outlined cell = lowest RMSE% in that window × moneyiness column. Grey = n = 0.

Analysis 2 · Does the ranking change with moneyness inside each window?

Table M6 · Best model (lowest RMSE%) by window × moneyness

Window	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	Modified GBM	Modified GBM	Modified GBM	Modified GBM	Modified GBM
2013-2014 Normal	—	Modified GBM	Modified GBM	Modified GBM	—
2018-2019 Late-cycle	—	Modified GBM	Modified GBM	Modified GBM	—
2019-2020 COVID	—	Modified GBM	Modified GBM	Modified GBM	—

Table M7 · n contracts by window × moneyness (shared across models)

Window	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	18	43	80	43	16
2013-2014 Normal	0	5	192	3	0
2018-2019 Late-cycle	0	4	191	7	0
2019-2020 COVID	0	6	189	5	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that window. Columns with n = 0 cannot change the ranking.

Does performance change with moneyness?

- DOTM (n = 18): BEST = Modified GBM with RMSE% = 43.64, Bias% = +18.53.
- OTM (n = 58): BEST = Modified GBM with RMSE% = 58.42, Bias% = +29.31.
- ATM (n = 652): BEST = Modified GBM with RMSE% = 46.30, Bias% = +27.92.
- ITM (n = 58): BEST = Modified GBM with RMSE% = 36.00, Bias% = +10.23.
- DITM (n = 16): BEST = Modified GBM with RMSE% = 31.39, Bias% = +10.29.
- Overall, the moneyness ranking is stable: Modified GBM is BEST in every non-empty bucket.
- Inside each regime, the BEST model is the same in every non-empty moneyness bucket.
- These comparisons use the Monday nearest-ATM sample, so ATM is the only well-populated bucket after 2009. DOTM/DITM results rest on few 2008–2009 quotes and should be read as descriptive, not as a high-powered test.